

Theorem 1.1. (The Division Theorem for $\mathbb{Z}$): $a, b \in \mathbb{Z}$ with $b > 0 \exists! q, r \in \mathbb{Z}$ with $0 \leq r < b$ s.t. $a = bq + r$.

Proof. (Existence): Let $R = \{a - bm : m \in \mathbb{Z} \text{ and } a - bm \geq 0\}$, then as: if $a \geq 0$, $a - b \cdot 0 = a \geq 0$ so $a \in R$; if $a < 0$, $a - b \cdot a = a(1 - b) \geq 0$ so $a - b \cdot a \in R$, $R \neq \emptyset$. Thus, take $r = \min(R)$ & so $0 \leq r = a - bm \in R$ for some $m \in \mathbb{Z}$, choose $q = m \in \mathbb{Z}$. So by construction $a = bq + r$. Note, $r < b$ as were such not true then $r > a - b(q + 1) = r - b \in R$ but this contradicts $r = \min(R)$, so we have $0 \leq r < b$. (Uniqueness): Suppose $q, q', r, r' \in \mathbb{Z}$ with $0 \leq r', r < b$ & $a = bq + r = bq' + r'$, also WLOG take $r' \leq r$. So with $\frac{r-r'}{b} \in (-1, 1)$, & $\frac{r-r'}{b} = \frac{a-bq-(a-bq')}{b} = q' - q \in \mathbb{Z}$, we have $\frac{r-r'}{b} = 0$ i.e. $r = r'$. So $q = \frac{a-r}{b} = \frac{a-r'}{b} = q'$. $\square$

Euclidian Algorithm:

Let $a, b \in \mathbb{Z}^+$, apply Th^m 1.1, to give:

$$\begin{aligned} a &= bq_1 + r_1 \\ b &= r_1q_2 + r_2 \\ r_1 &= r_2q_3 + r_3 \\ &\vdots \\ r_{n-1} &= r_nq_{n+1}, \end{aligned}$$

Here, $\gcd(a, b) = r_n$. To prove this relation we show: $G(r_i, r_{i+1}) = G(r_{i+1}, r_{i+2})$, where $G(a, b) = \{n \in \mathbb{Z} : n|a \text{ and } n|b\}$.

Reverse Euclid:

Given a completed algorithm we can obtain $n, m \in \mathbb{Z}$ s.t. $na + mb = \gcd(a, b)$ i.e. taking the start of a Euclidian algorithm:

$$\begin{aligned} a = q_1b + r_1 &\Rightarrow r_1 = a - q_1b \\ b = q_2r_1 + r_2 &\Rightarrow r_2 = b - q_2r_1 \\ &= b - q_2(a - q_1b) \\ &= -q_2a + (1 + q_1q_2)b \\ r_1 = q_3r_2 + r_3 &\Rightarrow r_3 = r_1 - q_3r_2 \\ &= (a - q_1b) \end{aligned}$$

as $a \neq 0$ & $b \neq 0$, respectively, the leading coefficient of $f g$ as $ab \neq 0$, hence $f g \neq 0$. $\square$

Proposition 3.26. Suppose I and J are ideals of the ring R . Then: (i) $I + J = \{a + b : a \in I, b \in J\}$ is an ideal; (ii)

Proof. Let $d = \gcd(f, g)$. Since $d \mid f$ and $d \mid g$, $\langle f, g \rangle \subseteq \langle d \rangle$. The Euclidean algorithm gives $d = af + bg$ for some

$a, b \in K[X]$, so $d \in \langle f, g \rangle$, hence $\langle d \rangle \subseteq \langle f, g \rangle$. □

Theorem 5.8. *Let K be a field. Then every ideal of $K[X]$ is principal.*

Proof. Let $I \triangleleft K[X]$. If $I = \{0\} = \langle 0 \rangle$, then I is principal. Otherwise choose non-zero $g \in I$ of least degree. Let $f \in I$. By Theorem 5.1, $f = qg + r$ with $r = 0$ or $\deg(r) < \deg(g)$. Since $r = f - qg \in I$, minimality of $\deg(g)$ forces $r = 0$. Hence every $f \in I$ is a multiple of g , so $I = \langle g \rangle$. □

Lemma 6.2. *Let K be a field and $f \in K[X]$ be irreducible. Then $\langle f \rangle$ is a maximal ideal of $K[X]$.*

Proof. (i) By Lemma 6.2, $\langle f \rangle$ is maximal, so L is a field by Corollary 4.13. Define $\theta: K \rightarrow L$ by $\theta(a) = a + \langle f \rangle$. This is a homomorphism. If $\theta(a) = \theta(b)$, then $a - b \in \langle f \rangle$. Since $\deg(f) \geq 1$, this forces $a - b = 0$, so θ is injective.

(ii): Let $f = b_0 + b_1X + \dots + b_nX^n$. For $\alpha = X + \langle f \rangle$, $f(\alpha) = b_0 + b_1\alpha + \dots + b_n\alpha^n = f + \langle f \rangle = \langle f \rangle = 0_L$. Thus α is a root of f in L .

(iii): If $g + \langle f \rangle \in L$, divide g by f : $g = qf + r$, with $r = 0$ or $\deg(r) < n$. Writing $r = a_0 + a_1X + \dots + a_{n-1}X^{n-1}$, we get $g + \langle f \rangle = r + \langle f \rangle = a_0 + a_1\alpha + \dots + a_{n-1}\alpha^{n-1}$. Thus L is spanned by $1, \alpha, \dots, \alpha^{n-1}$. Linear independence follows since any non-zero polynomial of degree $< n$ cannot lie in $\langle f \rangle$. □

Definitions

Definition 1.4. $\forall a, b \in \mathbb{Z}$, $\gcd(a, b) = \max\{n \in \mathbb{Z} : n|a \ \& \ n|b\}$, noting $\gcd(0, 0) = 0$ & if $\gcd(a, b) = 1$ then a & b are coprime.

Definition 1.6. $\forall a, b, n \in \mathbb{Z}$ with $n \geq 2$. a is **congruent to** b **mod** n **iff** $n|a - b$, written $a \equiv b \pmod n$. Then define the congruence class of a w.r.t. n as: $[a]_n = \{b \in \mathbb{Z} : b \equiv a \pmod n\}$

$\text{mod } n = \{b \in \mathbb{Z} : n \mid (b - a)\} = \{a + kn : k \in \mathbb{Z}\}$

We thus have:

$$\mathbb{Z}_n = \{[a]_n : a \in \mathbb{Z}\} = \{[0]_n, [1]_n, \dots, [n-1]_n\}$$

Note: $b \equiv a \pmod n$ iff $b = a + kn$ iff $[a]_n = \{a + nk : k \in \mathbb{Z}\}$ iff $[a]_n = [a + kn]_n$

Definition 1.14. A **ring** is a set R with two binary operations $+$ & $\times$, defined on R , satisfying the following:

- (R1) $(R, +)$ is an abelian group with identity 0.
- (R2) $\times$ is associative.
- (R3) $\times$ distributes over $+$ i.e.:

$$a(b + c) = ab + ac, \\ (a + b)c = ac + bc, \quad \forall a, b, c \in R$$

- (R4) The multiplicative identity $1 \in R$ & $1 \neq 0$

Definition 1.17. Let R be a ring & $S \subseteq R$. Then S is a **subring** of R if S is a ring w.r.t. the same addition & multiplication as R & contains the multiplicative identity of R , 1_R .

Definition 1.21. let R be a ring. The **ring of polynomials**

$R[X] = \{\sum_{i \geq 0} a_i X^i \mid \text{a finite sum} : a_i \in R\}$ with addition & multiplication defined as:

$$\sum_{i \geq 0} a_i X^i + \sum_{i \geq 0}^m b_i X^i = \sum_{i \geq 0}^{\max(n,m)} (a_i + b_i) X^i \ \& \ \\ (\sum_{i \geq 0} a_i X^i) (\sum_{i \geq 0}^m b_i X^i) = \sum_{i \geq 0}^{n+m} (\sum_{j=0}^i a_j b_{i-j}) X^i \text{ (taking missing coefficients as 0)}$$

$0 = \sum_{i \geq 0} 0X^i$ & $1 = 1X^0 + \sum_{i \geq 1} 0X^i$. Then the **degree** of a polynomial is the greatest $i \in \mathbb{N} \cup \{0\}$ s.t. $a_i \neq 0$ or if no such i exist $\deg(0) = -\infty$

More generally, we have:

$$R[X_1, \dots, X_n] = R[X_1, \dots, X_{n-1}][X_n]$$

$\forall f, g \in R[X], \deg(f + g) \leq \max(\deg(f), \deg(g))$ & $\deg(g) \leq \deg(f) + \deg(g)$.

Definition 1.24. If R_1 & R_2 are rings then the **Cartesian product** $R_1 \times R_2$ with the $+$ & $\times$ defined element wise is a ring.

Definition 2.1. For $a \in R$ define: for $n \in \mathbb{Z}^+$, $n \cdot a = \underbrace{a + a + \dots + a}_{n \text{ times}}$; for $n = 0$, $n \cdot a = 0 \cdot a = 0$; &

for $n \in \mathbb{Z}^-$, $n \cdot a = (-n) \cdot (-a)$.

The **Characteristic**, $\text{char}(R)$, of a ring R is the least positive integer n s.t. $n \cdot 1 = 0$, or if no exist $\text{char}(R) = 0$.

Note: $(n + m) \cdot a = n \cdot a + m \cdot a$, $n \cdot (a + b) = n \cdot a + n \cdot b$, & $n \cdot (m \cdot a) = (nm) \cdot a$, all easily verifiable.

Definition 2.3. A non-zero element $r \in R$ is a **zero-divisor** if $\exists s \in R$ a non-zero element s.t. $rs = 0$ or $sr = 0$.

A ring R is a domain if, $\forall r, s \in R$:

$$rs = 0 \implies r = 0 \text{ or } s = 0.$$

i.e. a ring with no zero divisors is a domain & vice versa.

Definition 2.9. A **division ring** is a ring where every non-zero element has a right & left inverse, i.e.: $\forall r \in R \setminus \{0\}, \exists s, t \in R$ s.t. $rs = 1 = tr$ with s, t being the right & left inverses, respectively. Note: this immediately collapse, due to Le^m 2.11, to just there exists an inverse as $s = t$.

An **invertible** element r is called a **unit**.

Definition 2.16. An element r of a ring R is **nilpotent** if $\exists n \in \mathbb{Z}^+$ s.t. $r^n = 0$, and the least such n is the **index of nilpotence** of r .

An element $r \in R$ is **idempotent** if $r^2 = r$.

Note: In any ring R , 0_R is nilpotent with index 1 and 0_R and 1_R are idempotent.

Definition 3.1. If R & S are rings then an **isomorphism** from R to S is a bijection $\phi: R \rightarrow S$ s.t. $\forall r, r' \in R$:

$$\phi(r + r') = \phi(r) + \phi(r'),$$

$$\phi(rr') = \phi(r)\phi(r')$$

$\exists \phi: R \rightarrow S$ an isomorphism iff R & S are **isomorphic**, written $R \simeq S$.

Definition 3.6. If R & S are rings then a **homomorphism** from R to S is a map $\vartheta: R \rightarrow S$ s.t. $\forall r, r' \in R$:

$$\vartheta(r + r') = \vartheta(r) + \vartheta(r'),$$

$$\vartheta(rr') = \vartheta(r)\vartheta(r'),$$

$$\vartheta(1_R) = 1_S.$$

Definition 3.9. An **embedding** or **monomorphism** is an injective homomorphism. There is a trivial embedding along each arrow of $\mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R} \rightarrow \mathbb{C} \rightarrow \mathbb{C}[X]$.

Definition 3.12. If $\vartheta: R \rightarrow S$ is a ring homomorphism, then the **kernel** of ϑ is defined as $\ker(\vartheta) = \{r \in R : \vartheta(r) = 0_S\}$.

Definition 3.16. An **automorphism** of a ring is an isomorphism from and to itself.

Definition 3.18. An **ideal** of a ring R is a subset $I \subseteq R$ s.t.: $0 \in I$; $a + b \in I, \forall a, b \in I$; & $ar, ra \in I, \forall a \in I, \forall r \in R$. We write $I \triangleleft R$ to mean I is an ideal of R .

Definition 3.19. If $a \in R$ then $\{r_1as_1 + \dots + r_nas_n : n \geq 1, r_i, s_i \in R\}$ is an ideal containing a and is the smallest ideal of R containing a . It is called the **principal ideal generated by** a ,

denoted $\langle a \rangle$. If R is commutative then

$\langle a \rangle = \{ar : r \in R\}$. A **principal ideal** is one which can be generated by a single element.

Definition 3.21. A **right ideal** is defined as for a general ideal but with the third condition replaced by the weaker condition: $a \in I$ and $r \in R$ implies $ar \in I$. A **left ideal** is defined similarly. If $a \in R$, the

principal right ideal generated by a is $\{ar : r \in R\}$, denoted aR .

Definition 3.22. Let $A \subseteq R$ and R be a commutative ring, then $\{a_1r_1 + \dots + a_nr_n : n \geq 1, a_i \in A, r_i \in R\}$ is the **ideal generated by** A .

Definition 3.26. We can also define a **product of ideals**: if $I, J \triangleleft R$, then their product is the ideal generated by $\{ab : a \in I, b \in J\}$. This set will not in general be closed under addition, hence we say “generated by”. Powers of an ideal are defined inductively by $I^{n+1} = I^nI$.

Definition 4.1. Let R be a ring and let I be a proper ideal. Let R/I denote the set of cosets of I in the additive group $(R, +)$:

$$R/I = \{r + I : r \in R\}.$$

Define operations $+$ and $\times$ on R/I by $(r + I) + (s + I) = (r + s) + I, \quad (r + I)(s + I) = rs + I$.

Definition 4.1. The ring R/I is called the **factor ring** (or **quotient ring**) of R by I .

Exercise. Write out the elements of the factor ring $\mathbb{Z}_6/\langle[3]_6\rangle$. What is this factor ring isomorphic to?

Definition 4.4. The map $\pi: R \rightarrow R/I, \pi(r) = r + I$, is called the **canonical projection**, or **canonical surjection**.

Notation: if $I \subseteq J$ are ideals then usually write $I \leq J$ (or $J \geq I$) instead of subset notation, with the same meaning.

The notation J/I is also used instead of πJ for the image of J in R/I .

Definition 4.12. An ideal I of a ring R is **maximal** if it is proper and for any ideal J with $I \leq J \leq R$, either $J = I$ or $J = R$.

Definition 4.14. A proper ideal I of a commutative ring R is **prime** if whenever $r, s \in R$ and $rs \in I$, then $r \in I$ or $s \in I$.

Definition 5.3. An element $a \in K$ is a **root** (or **zero**) of $f \in K[X]$ iff $f(a) = 0$.

Definition 5.5. The **greatest common divisor** of polynomials f, g is a polynomial d such that $d \mid f$ and $d \mid g$, and if $h \mid f$ and $h \mid g$, then $h \mid d$. This determines d up to a non-zero scalar; for uniqueness take d **monic**.

Division Algorithm for Polynomials. For $f, g \in K[X], g \neq 0$, repeatedly apply Th^m 5.1:

$$f = q_1g + r_1, \quad \deg(r_1) < \deg(g), \\ g = q_2r_1 + r_2, \quad \deg(r_2) < \deg(r_1), \\ r_1 = q_3r_2 + r_3, \quad \deg(r_3) < \deg(r_2), \\ \vdots \\ r_{n-1} = q_{n-1}r_n.$$

Since degrees strictly decrease, some $r_{n+1} = 0$. Then $\gcd(f, g) = \gcd(g, r_1) = \dots = \gcd(r_{n-1}, r_n) = r_n$ up to multiplication by a non-zero scalar; take the monic version. Tracking back expresses $\gcd(f, g) = kf + lg$ for some $k, l \in K[X]$.

Factorisation. Here R is commutative.

Definition 5.9. (i) $r \in R$ is **irreducible** if r is not invertible and whenever $r = st$, either s or t is invertible. (ii) $r, s \in R$ are **associated** if $s = ur$ for some invertible $u \in R$.

Definition 5.11. A commutative domain R is a **Unique Factorisation Domain (UFD)** if every non-zero, non-invertible element factors uniquely into irreducibles, up to rearrangement and associated factors.

Definition 5.14. A **Principal Ideal Domain (PID)** is a commutative domain in which every ideal is principal. To produce a root for a polynomial p , it is enough to produce a root of one irreducible factor of p . Hence focus on irreducible polynomials.

Examples

Example. With $n \geq 2$, $M_n(R)$ is the ring of $n \times n$ matrices, a non-commutative ring. Note: the only two-sided ideals of $M_n(F)$, F a field, are $M_n(F)$ & $\{0\}$ but this isn't a division ring hence even relaxing commutative in Pro^* 3.22 doesn't yeild the same pro^* with field $\rightarrow$ division ring.

Example. For X a set $\mathcal{P}(X)$ with $A + B = (A \cup B) \setminus (A \cap B),$ $AB = A \cap B$ & $0 = \emptyset, -A = A, A, B \in \mathcal{P}(X)$.

Example. Division rings are not all fields e.g. the quaternions $\mathbb{H} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{R}\}$ where $i^2 = j^2 = k^2 = -1$ & $ij = k, jk = i, \& \ ki = j$, this is non-commutative as $-ij = -k = jjk = ji$ so $ij \neq ji$ and this is a division ring as for $z = a + bi + cj + dk$ $z^{-1} = \frac{-a-bi-cj-dk}{a^2+b^2+c^2+d^2}$, taking $z \neq 0$.

Example. For 4 element rings $\mathbb{Z}_4 \not\simeq \mathbb{Z}_2 \times \mathbb{Z}_2$, since given $[2]_4 \neq 0$ and $[2]_4^2 = [0]_4$, $[2]_4$ is nilpotent, where as there are no non-zero nilpotent elements in $\mathbb{Z}_2 \times \mathbb{Z}_2$, hence the result by Le^m 3.3.

Example. The map $\mathbb{Z} \rightarrow \mathbb{Z}$ by $m \mapsto [m]_n$ is a homomorphism. Also, $\vartheta: \mathbb{Z}_{kn} \rightarrow \mathbb{Z}_n$ by $\vartheta([m]_{kn}) = [m]_n$ is a homomorphism, well definedness must be shown, and is true as; let $[m]_{kn} = [m]_{kn} \in \mathbb{Z}_{kn}$ then $kn \mid m - m'$ so $n \mid m - m'$ thus $[m]_n = [m']_n$.

Example. With R a ring, the map $R[X] \rightarrow R[X]$ by $p(X) \mapsto p(q(X))$ for some $q(X) \in R[X]$.

Example. $\mathbb{Z}[i] \rightarrow \mathbb{Z}[i]$ by $a + bi \mapsto a - bi$ is an automorphism.

Example. Examples 3.20. (i) In every ring $\langle 0 \rangle = \{0\}$ is the smallest ideal, the **trivial ideal**. (ii) In every ring $\langle 1 \rangle = R$ is an ideal and every other ideal $\neq R$ is a **proper ideal**. (iii) If $n \in \mathbb{Z}$ then $\langle n \rangle = \{nk : k \in \mathbb{Z}\}$. (iv) If $p \in R[X]$ then $\langle p \rangle$ is the set of polynomials with p as a factor.

Example. Let $R = M_2(\mathbb{Z})$ and $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ then by simple multiplication the principle left and right ideals are $\{\begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix} : a, b \in \mathbb{Z}\}$ and $\{\begin{pmatrix} a & b \\ 0 & 0 \end{pmatrix} : a, b \in \mathbb{Z}\}$ whilst

$$\langle A \rangle = M_2(\mathbb{Z}) \text{ as } I_2 = A + \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} A \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \in \langle A \rangle$$

Note: If $1 \in I$ with $I \triangleleft R$ the $\forall r \in R$ $r = r1 \in I$, so $I = R$.

Example. Working in $\mathbb{Z}[X]$ the ideal $I = \langle 2, X \rangle := \langle \{2, X\} \rangle$ is not a principal ideal. Suppose $I = \langle p \rangle$. Since $2 \in I$, $2 = pq$ for some $q \in \mathbb{Z}[X]$. As $\mathbb{Z}$ is a domain, $0 = \deg(2) = \deg(pq) = \deg(p) + \deg(q)$, so $p, q \in \mathbb{Z}$ and $p = \pm 1$ or $p = \pm 2$. If $p = \pm 1$, then $1 = 2r_1 + Xr_2$ for some $r_1, r_2 \in \mathbb{Z}[X]$, impossible since the RHS has even constant term. Hence $p = \pm 2$. But $X \in I$, so $X = 2r_3$ for some $r_3 \in \mathbb{Z}[X]$, impossible since $2r_3$ has even coefficients. Thus I is not principal.

Example. $\mathbb{Z}/\langle n \rangle = \mathbb{Z}_n$.

Example. 4.6. Take $R = \mathbb{Z}$ and $I = \langle 6 \rangle = 6\mathbb{Z}$, so the canonical projection is $\pi: \mathbb{Z} \rightarrow \mathbb{Z}_6$. Let $\vartheta: \mathbb{Z} \rightarrow \mathbb{Z}/\langle 2 \rangle = \mathbb{Z}_2$ be the canonical projection. Then $\ker(\vartheta) = \langle 2 \rangle \supseteq \langle 6 \rangle$, so by Le^m 4.4 there is a unique homomorphism ϑ' with $\vartheta' \circ \pi = \vartheta$. This is the map $\vartheta': \mathbb{Z}_6 \rightarrow \mathbb{Z}_2$ already seen in Examples 3.8.

Example. 4.7. Let $R = \mathbb{R}[X]$ and $I = \langle X^2 + 1 \rangle$. Let $\pi: \mathbb{R}[X] \rightarrow \mathbb{R}[X]/\langle X^2 + 1 \rangle$ be the canonical surjection. Let $\vartheta: \mathbb{R}[X] \rightarrow \mathbb{C}$ be evaluation at i , $\vartheta(p(X)) = p(i)$. Then $p \in \ker(\vartheta)$ iff $X^2 + 1 \mid p$, so $\ker(\vartheta) = \langle X^2 + 1 \rangle$. By Le^m 4.4, there is a unique map $\vartheta': \mathbb{R}[X]/\langle X^2 + 1 \rangle \rightarrow \mathbb{C}$ through which ϑ factors. It is injective and surjective, hence an isomorphism. Thus $\mathbb{R}[X]/\langle X^2 + 1 \rangle \simeq \mathbb{C}$.

Example. 4.9. The ideals of $\mathbb{Z}_6 = \mathbb{Z}/\langle 6 \rangle$ are in one-to-one correspondence with the ideals of $\mathbb{Z}$ containing $\langle 6 \rangle$. There are 4 of each: $\langle 1 \rangle, \langle 2 \rangle, \langle 3 \rangle, \langle 6 \rangle$ correspond to

$$\pi(1) = \mathbb{Z}_6 = \{[0]_6, [1]_6, [2]_6, [3]_6, [4]_6, [5]_6\},$$

$$\pi(2) = \langle [2]_6 \rangle = \{[0]_6, [2]_6, [4]_6\},$$

$$\pi(3) = \langle [3]_6 \rangle = \{[0]_6, [3]_6\},$$

$$\pi(6) = \langle [0]_6 \rangle = \{[0]_6\}.$$

Example. 4.11. Since $\langle 6 \rangle \leq \langle 2 \rangle$, Th^m 4.10 gives $(\mathbb{Z}/\langle 6 \rangle)/(\langle 2 \rangle/\langle 6 \rangle) \simeq \mathbb{Z}/\langle 2 \rangle$.

Example. 4.15. All ideals of $\mathbb{Z}$ are principal. The prime ideals are precisely the ideals generated by a prime number or zero.

Example. 5.2. In $\mathbb{Z}_6[X]$, $X^4 + 2X^2 + X + 4 = (X^2 + 2X + 3)(X^2 + 3X + 3) + X$.

Example. Examples 5.10. (i) In $\mathbb{Z}$, irreducibles are $\pm p$ for p prime; r, s are associated iff $r = \pm s$. (ii) In $\mathbb{R}[X]$, irreducibles are non-constant polynomials with no factor g satisfying $1 \leq \deg(g) < \deg(f)$.

Example. 5.12. $\mathbb{Z}$ is a UFD. For example, $-24 = (-2) \cdot 2 \cdot (-2) \cdot (-3)$, unique up to signs and ordering.

Example. 5.13. $\mathbb{Z}[\sqrt{-5}]$ is not a UFD since $6 = 3 \cdot 2 = (1 + \sqrt{-5})(1 - \sqrt{-5})$, with irreducible factors not matching up to associates.

Example. Examples 5.15. $\mathbb{Z}$ is a PID; if K is a field, $K[X]$ is a PID by Th^m 5.8. Every PID is a UFD. If R is a UFD then so is $R[X]$, hence $R[X_1, \dots, X_n]$.

Example. $\mathbb{Z}[X]$ is a UFD but not a PID: it is a UFD since $\mathbb{Z}$ is, but $\langle 2, X \rangle$ is not principal.

Example. Examples 5.16. (i) $X^3 + X^2 + 1$ is irreducible in $\mathbb{Z}_2[X]$: a reducible cubic has a linear factor, but 0, 1 are not roots. (ii) $X^4 + 2X^3 + X + 1$ is irreducible in $\mathbb{Z}_3[X]$: no roots, and comparing coefficients in two possible quadratics gives no factorisation.

Example. 6.1. Take $p = X^2 - 2 \in \mathbb{Q}[X]$. An extension field containing a root is $\mathbb{R}$, or more efficiently $\mathbb{Q}[\sqrt{2}]$.

Example. 6.4. (i) If $K = \mathbb{Q}$ and $f = X^2 - 2$, then $\mathbb{Q}[X]/\langle X^2 - 2 \rangle \simeq \mathbb{Q}[\sqrt{2}]$. (ii) If $K = \mathbb{R}$ and $f = X^2 + 1$, every element of $\mathbb{R}[X]/\langle X^2 + 1 \rangle$ has form $a_0 + a_1i$, using i for $X + \langle f \rangle$, so $\mathbb{R}[X]/\langle X^2 + 1 \rangle \simeq \mathbb{C}$.